

The **Document Object Model (DOM)** is a programming interface for HTML and XML documents. JavaScript can access and change the page's content, structure, and style.

JavaScript uses the DOM to:

- Access HTML elements
- Change text and content
- Change CSS styles
- Add or remove elements
- Handle events like clicks

DOM Basics

1. Selecting Elements

getElementById()

Selects an element by its ID.

```
const heading = document.getElementById("title");  
console.log(heading);
```

getElementsByClassName()

Returns all elements with the given class.

```
const items = document.getElementsByClassName("text");  
console.log(items);
```

getElementsByTagName()

Returns all elements of a specific tag.

```
const paragraphs = document.getElementsByTagName("p");
```

querySelector()

Returns the **first matching element**.

```
const heading = document.querySelector("#title");
```

Examples

By ID

```
document.querySelector("#title");
```

By Class

```
document.querySelector(".text");
```

By Tag

```
document.querySelector("p");
```

querySelectorAll()

Returns **all matching elements** as a NodeList.

HTML

```
<p>One</p>
```

```
<p>Two</p>
```

```
<p>Three</p>
```

JavaScript

```
const paragraphs = document.querySelectorAll("p");
```

```
paragraphs.forEach(function(item){  
    console.log(item.textContent);  
});
```

Output

One

Two

Three

Changing Content

innerHTML

Gets or sets the HTML inside an element.

HTML

```
<p id="demo">Hello</p>
```

JavaScript

```
document.getElementById("demo").innerHTML = "<b>Welcome</b>";
```

Output

```
<b>Welcome</b>
```

textContent

Changes only the text.

```
document.getElementById("demo").textContent = "JavaScript";
```

Output

JavaScript

HTML tags are shown as plain text, not interpreted.

innerText

Returns only the visible text.

```
document.getElementById("demo").innerText = "Visible Text";
```

Changing Styles

```
const heading = document.querySelector("h1");
```

```
heading.style.color = "blue";
```

```
heading.style.fontSize = "40px";
```

Changing Attributes

getAttribute()

```
const img = document.querySelector("img");
```

```
console.log(img.getAttribute("src"));
```

setAttribute()

```
img.setAttribute("src", "flower.jpg");
```

Creating Elements

```
const newPara = document.createElement("p");
```

```
newPara.textContent = "New Paragraph";
```

Adding Elements

```
document.body.appendChild(newPara);
```

Removing Elements

```
const para = document.querySelector("p");
```

```
para.remove();
```

Event Handling

HTML

```
<button id="btn">Click Me</button>
```

JavaScript

```
document.getElementById("btn").addEventListener("click", function(){  
    alert("Button Clicked!");  
});
```

Difference Between `querySelector()` and `querySelectorAll()`

Feature	<code>querySelector()</code>	<code>querySelectorAll()</code>
Returns	First matching element	All matching elements
Return Type	Element	NodeList
CSS Selectors	Yes	Yes

Example

```
document.querySelector(".item");
```

Returns only the first `.item`.

```
document.querySelectorAll(".item");
```

Returns all `.item` elements.

Difference Between `innerHTML`, `innerText`, and `textContent`

Property	HTML Tags	Hidden Text	Purpose
innerHTML	Interprets HTML	Yes	Get/set HTML content
innerText	No	No	Get/set visible text
textContent	No	Yes	Get/set all text content

Example

```
<p id="demo">Hello <b>World</b></p>
```

```
const p = document.getElementById("demo");
```

```
console.log(p.innerHTML); // Hello <b>World</b>
```

```
console.log(p.innerText); // Hello World
```

```
console.log(p.textContent); // Hello World
```
